

GALENA, AK, USA

WMO: 702220

Lat: 64.737N

Lon: 156.934W

Elev: 153

StdP: 14.61

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26501

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.0	-41.2	-44.7	0.4	-36.0	-39.3	0.6	-31.0	23.6	16.2	20.4	16.7	0.9	90	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	78.6	60.8	74.9	59.1	71.9	58.0	63.0	73.6	61.2	71.4	59.5	68.8	6.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.8	74.3	65.6	56.9	69.3	64.2	55.2	65.3	62.7	28.5	73.4	27.2	71.3	26.1	69.0	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	16.9	14.3	DB	-48.9	83.5	4.8	3.2	-52.3	85.8	-55.1	87.6	-57.8	89.4	-61.2	91.7	(4)
(5)			WB	-37.4	65.8	4.7	2.5	-40.7	67.5	-43.4	69.0	-46.1	70.3	-49.4	72.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	-7.4	3.6	8.6	28.0	46.8	58.2	60.4	54.6	44.5	27.1	5.1	-3.6	(6)
(7)		DBStd	27.31	21.12	16.30	13.90	11.32	8.11	5.84	5.08	5.74	7.21	10.96	15.59	16.87	(7)
(8)		HDD50	9113	1779	1299	1282	659	154	7	2	20	189	711	1348	1663	(8)
(9)		HDD65	13797	2244	1719	1747	1109	566	215	156	324	616	1175	1798	2128	(9)
(10)		CDD50	818	0	0	0	0	53	253	325	164	23	0	0	0	(10)
(11)		CDD65	27	0	0	0	0	0	10	15	2	0	0	0	0	(11)
(12)		CDH74	378	0	0	0	0	10	142	180	46	0	0	0	0	(12)
(13)		CDH80	49	0	0	0	0	1	19	24	5	0	0	0	0	(13)
(14)	Wind	WSAvg	5.2	4.0	5.0	5.3	5.9	5.7	5.4	5.8	5.6	5.5	5.6	4.8	4.1	(14)
(15)	Precipitation	PrecAvg	12.8	0.7	0.7	0.6	0.4	0.5	1.4	2.0	2.4	1.6	1.0	0.8	0.9	(15)
(16)		PrecMax	18.2	1.8	1.7	1.5	1.1	0.9	2.8	4.2	4.0	3.8	2.1	2.3	2.4	(16)
(17)		PrecMin	9.6	0.0	0.0	0.0	0.0	0.0	0.2	0.5	0.8	0.5	0.0	0.0	0.2	(17)
(18)		PrecStd	2.3	0.4	0.5	0.4	0.4	0.3	0.8	0.8	0.8	0.9	0.6	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.0	37.4	40.4	57.9	75.4	82.3	83.4	79.8	67.7	51.6	38.0	35.1	(19)
(20)			MCWB	34.4	34.6	34.2	44.4	55.2	61.7	63.4	63.4	54.6	45.6	35.0	32.5	(20)
(21)		2%	DB	32.5	34.6	36.4	52.4	70.2	78.1	79.0	73.2	63.0	45.9	35.0	31.0	(21)
(22)			MCWB	30.3	31.7	31.9	41.6	52.8	59.3	61.5	60.1	51.6	41.4	32.8	29.0	(22)
(23)		5%	DB	27.0	30.9	33.0	48.3	65.9	74.6	75.2	69.3	59.3	43.2	31.7	25.2	(23)
(24)			MCWB	25.4	28.5	29.2	39.3	50.8	57.9	60.4	58.7	50.6	40.1	30.0	23.6	(24)
(25)		10%	DB	21.3	26.4	28.9	44.8	61.8	71.5	72.2	65.1	55.9	40.2	26.7	18.9	(25)
(26)			MCWB	20.1	24.3	25.7	37.6	48.9	56.6	59.3	56.8	49.3	37.7	25.3	18.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.4	35.4	35.7	45.3	56.7	64.3	65.5	65.4	55.5	46.4	35.5	32.8	(27)
(28)			MCDWB	37.2	37.0	38.9	56.6	71.8	76.8	79.2	76.0	64.0	49.1	37.0	34.9	(28)
(29)		2%	WB	31.0	32.1	32.6	42.3	54.3	61.7	63.4	62.0	53.2	42.3	33.1	29.6	(29)
(30)			MCDWB	33.0	34.4	35.9	50.9	67.5	73.6	74.4	70.4	60.5	44.8	34.8	31.4	(30)
(31)		5%	WB	26.5	28.6	29.1	40.1	52.0	59.6	61.7	59.8	51.4	40.3	29.9	24.1	(31)
(32)			MCDWB	28.2	31.1	32.6	47.5	63.3	71.0	72.5	67.1	57.7	42.9	31.8	25.9	(32)
(33)		10%	WB	21.6	24.5	25.8	37.7	50.2	58.0	60.1	57.9	50.0	37.8	25.5	18.7	(33)
(34)			MCDWB	22.7	26.5	29.1	44.2	60.0	68.5	69.9	63.7	55.3	40.2	26.9	19.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.1	18.3	22.8	20.6	19.8	18.8	16.4	15.8	15.0	11.9	13.5	13.7	(35)
(36)			MCDBR	13.2	13.7	20.8	22.8	26.8	25.7	24.2	24.6	22.6	12.1	10.6	14.1	(36)
(37)			MCWBR	11.9	11.9	16.5	13.6	12.7	10.5	10.3	12.5	13.2	8.7	9.3	12.9	(37)
(38)		5% WB	MCDWB	13.3	14.0	18.5	20.8	23.6	21.7	21.2	19.7	18.6	10.8	10.5	13.9	(38)
(39)			MCWBR	12.1	12.5	14.7	12.8	11.4	9.6	9.5	10.3	11.1	8.3	9.3	12.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.172	0.247	0.257	0.339	0.340	0.365	0.366	0.368	0.307	0.279	N/A	0.176	(40)	
(41)		taud	1.937	2.206	2.283	2.122	2.406	2.371	2.410	2.417	2.558	2.396	N/A	1.931	(41)	
(42)		Ebn at Noon	122	213	268	260	276	269	266	253	251	201	N/A	39	(42)	
(43)		Edh at Noon	7	17	25	38	31	33	31	29	20	15	N/A	3	(43)	
(44)	All-Sky Solar Radiation	RadAvg	51	261	848	1487	1713	1769	1484	1109	705	280	77	15	(44)	
(45)		RadStd	7	28	65	102	183	148	109	113	84	40	11	1	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page